Edoardo Maggioni

Computational Social Scientist | Network Science

website edoarder edo.magg@me.com +39 3913349580



Research Interests

- Collective behaviour and opinion dynamics
- Information diffusion and online communication
- Social simulation and agent-based modelling
- Environmental communication and behaviour

Experience

IT UNIVERSITY OF COPENHAGEN - NERDS | Internship

October 2024 – June 2025 | Copenhagen, Denmark

- Integrated social media data, Eurobarometer surveys and Eurostat indicators; used statistical modelling, robustness checks, visualisation and NLP to study pro-environmental behaviour across European regions (Research Square preprint).

ETH ZÜRICH - SYSTEMS DESIGN | Internship

February 2021 – September 2021 | Zürich, Switzerland

- Developed an agent-based model that generates social interaction patterns from signed relations, incorporating structural balance and geometrical frustration; implemented simulations, parameter exploration and network-level analysis in Python.

NATECH S.R.L. | Scientific Collaboration

June 2018 – February 2022 | Siena, Italy

- Contributed to an R&D project (Regione Toscana) through theoretical and computational analysis; developed a prototype ultrasound phased-array device.
- Participated in drafting material for an EU Horizon 2020 call.

CERN | Internship

August 2017 – September 2017 | Geneva, Switzerland

- Data collection and sensor analysis for timing detectors at the TOTEM experiment.

Education

UNIVERSITY OF PISA | National PhD in AI for Society

October 2022 – Current | Lucca and Pisa, Italy

Thesis officially approved; defence scheduled for 13 November 2026

- *Networks in Society: Complex Systems Approaches to Digital Communication and Environmental Change*; supervisors: Rossana Mastrandrea and Diego Garlaschelli.
- Research at IMT Lucca and IIT-CNR on collective behaviour and online communication, combining social-media traces, Eurobarometer surveys, official statistics and network models.
- Extended a random-walk motif approach to temporal, directed social-media networks (arXiv:2605.30551).

UNIVERSITY OF PISA | Master's in Theoretical Physics

September 2018 - October 2021 | Pisa, Italy

Final grade: 110 / 110

UNIVERSITY OF SIENA | Bachelor's in Physical Sciences and Advanced Technologies

October 2014 - April 2018 | Siena, Italy

Final grade: 110 / 110 cum Laude

Skills

PROGRAMMING

Python • Git/GitHub
LaTeX • MATLAB
NumPy • pandas
SciPy • StatsModels
NetworkX • Mesa

METHODS

- Statistical modelling, robustness and sensitivity
- Network reconstruction, null models and validation
- Temporal and multilayer network analysis
- Agent-based simulation and parameter exploration
- Data integration, NLP and scientific visualisation

LANGUAGES

- Italian: Native
- English: C1
- French, German: A1

Interests

Curiosity, creativity and originality are important to me, both in research and in life. I enjoy music, meteorology and travelling, and I practised judo for ten years.

References

Prof. Diego Garlaschelli
IMT Lucca, NETWORKS
diego.garlaschelli@imtlucca.it

Prof. Luca Maria Aiello
IT University of Copenhagen
luai@itu.dk

Dr. Rossana Mastrandrea
University of Turin
rossana.mastrandrea@unito.it

Edoardo Maggioni

Publications and Research Outputs

Three research manuscripts under journal consideration or revision; two are publicly available as preprints.

Twitter climate discourse as a signal of pro-environmental behaviors

(E. Maggioni, D. Garlaschelli, R. Mastrandrea, L. M. Aiello)

Major revision requested by EPJ Data Science; Research Square preprint: 10.21203/rs.3.rs-10091583/v1

Cross-regional analysis linking climate-related Twitter discourse to Eurobarometer measures of pro-environmental behaviour and regional socio-economic indicators from Eurostat. Combines statistical modelling, robustness and sensitivity checks, and NLP-based dimensions of communication to study relationships between online discourse and self-reported behaviour.

Digital communication infrastructures in Irish offshore wind governance: A multilayer network analysis

(E. Maggioni, N. Ludovico, M. Esteve del Valle, F. Ruzzenenti, R. Mastrandrea)

Under review at Energy Research & Social Science; no public preprint currently available

Multi-layer analysis of the Irish offshore-wind governance ecosystem, combining institutional hyperlink and Twitter interaction networks among public bodies, companies, associations, NGOs and research organisations. Examines fragmentation, coordination, brokerage and the emergence of persistent relational structures across communication environments.

Beyond Direct Retweets: Multi-Step Pathways in Italian COVID-19 Twitter

(E. Maggioni, R. Manfredi, F. Saracco, R. Mastrandrea)

Joint first authorship with R. Manfredi.

Submitted to Social Networks; arXiv preprint: 10.48550/arXiv.2605.30551

Analysis of Italian COVID-19 retweet networks using random walks and motif-based methods to study persistence, spillovers and cross-community trajectories beyond direct retweets. Extends a random-walk motif framework to directed, temporal communication networks.

Selected Conference Contributions

- **Conference on Complex Systems (CCS), Siena, Italy — 2025** — Oral presentation
- **NetSci, Maastricht, Netherlands — 2025** — Satellite talk
- **International Conference on Computational Social Science (IC2S2), Norrköping, Sweden — 2025** — Poster
- **StatPhys, Florence, Italy — 2025** — Poster
- **International AAAI Conference on Web and Social Media (ICWSM), Copenhagen, Denmark — 2025** — Satellite talk

References

Prof. Diego Garlaschelli
Director, NETWORKS Research Unit
IMT Lucca
diego.garlaschelli@imtlucca.it

Prof. Luca Maria Aiello
Full Professor of Data Science
IT University of Copenhagen
luai@itu.dk

Dr. Rossana Mastrandrea
Tenure Track Researcher
Department of Management
"Valter Cantino", University of Turin
rossana.mastrandrea@unito.it